

# WG21 2018-11 San Diego

## Record of Discussion

ISO/IEC JTC1 SC22 WG21 P1338R1 — 2018-11-26

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Chair John Spicer

Changes from previous revision:

- Fixing various typos
- Amending time zone details

### 1. Opening activities

John Spicer opened the meeting at 9:05 PST.

#### 1.1 Opening comments, welcome from host

Marshall Clow welcomes the group and presents the information about the location of the meeting. Lunch is provided at the hotel by Qualcomm  
Thank you to Qualcomm.

John Spicer presents the schedule for the meeting. Working groups will meet until Friday. Plenary is scheduled for Saturday at 8:30 am PST. There are several evening sessions scheduled. Some subgroups may have a shorter lunch period, please check with your group chair.

C++ foundation meeting will happen this evening in the plenary room. The attendance is optional.

If you have any questions, the schedule can be found on the wiki

Marshall: If you are new, please ask the person next to you for information on how to access the meeting wiki.

#### 1.2 Meeting Guidelines

John Spicer presents.

Every participant is responsible for understanding and abiding by the following:

- [The INCITS Antitrust Guidelines](#) (PL22.16)
- [The INCITS Patent Policy](#) (PL22.16)
- [The ISO Code of Conduct](#)
- [The WG21 Practices and Procedures, and Code of Conduct](#)

Please review the guidelines. WG21 procedures and Code of Conduct documents are new and very important. We want to make sure that everybody interacts in a professional and respectful way.

If you have any questions talk to one of the officials.

## 1.3 Membership, voting rights, and procedures for the meeting

John Spicer presents. Meetings are not public, but we allow visitors. Speak freely, don't do live tweeting, don't quote people without their permissions. Photography is fine, please post after the meeting.

For the US organization 1 vote per company, for other countries, one vote per national expert.

In the WG and SG everyone can take part in straw polls

For more information about membership talk to Hal Finkel.

Barry Hedquist presents. PL22.16 has the ballot process on Saturday. The Saturday vote is approval of Systematic Review for ISO/IEC TS 19217:2015 -- C++ Extensions for Concepts. If you are not here on Saturday, and you are a voting member, please send an email to John, Hal or Barry.

## 1.4 Introductions

Officers and WG chairs introduce themselves

First time attendees introduce themselves.

John Spicer welcomes first time attendees.

## 1.5 Agenda review and approval

John Spicer presents the agenda for the meeting. The meeting will finish no later than 2pm on Saturday, but WG may continue working.

Barry Hedquist moves. Adam Martin seconds.  
The motion is unanimously approved by PL22/16.  
The motion is unanimously approved by WG21.

## 1.6 Editor's reports, approval of working drafts

| Document                | Editor's report       | Prospective WD        |
|-------------------------|-----------------------|-----------------------|
| C++20 Standard          | <a href="#">N4779</a> | <a href="#">N4778</a> |
| Coroutines TS           | <a href="#">N4776</a> | <a href="#">N4775</a> |
| Networking TS           | <a href="#">N4772</a> | <a href="#">N4771</a> |
| Parallelism v2 TS       | <a href="#">DONE</a>  | <a href="#">DONE</a>  |
| Reflection TS           | <a href="#">N4767</a> | <a href="#">N4766</a> |
| Library Fundamentals TS | <a href="#">N4759</a> | <a href="#">N4758</a> |

Motion to approve editor's reports and working drafts.  
The motion is unanimously approved by WG21.

## 1.7 Approval of the minutes of the previous meetings

| Meeting                                   | Minutes                            |
|-------------------------------------------|------------------------------------|
| WG21 Rapperswil                           | <a href="#">N4781</a>              |
| PL22.16 Rapperswil                        | <a href="#">pl22.16-2018-00010</a> |
| WG21 pre-San Diego administrative telecon | <a href="#">N4784</a>              |

Motion to approve the minutes of the previous meeting.

The motion is unanimously approved by WG21.

PL22/16 motion to approve the minutes of the previous meeting.

Marshall Clow moves, Barry Hedquist seconds.

The motion is unanimously approved by PL22/16.

The motion is unanimously approved by WG21.

## 2. Liaison reports, and WG21 study group reports (see pre-meeting WG21 telecon minutes)

No discussion.

## 3. WG progress reports and work plans for the week (Core, Evolution, Library, Library Evolution; see pre-meeting WG21 telecon minutes)

No discussion.

## 4. New business requiring action by the committee

Herb Sutter presents : There are 181 people at this meeting. We have grown a lot and the number of papers has grown a lot too. We used to have less papers than people, but that is no longer the case. We still want to look at all of the papers

We have 12 NB, next year we may have 14 NB

To be able to see all the papers, we have scaled the groups and added incubation groups at the meeting. We are still bottlenecked at LWG and CWG which need to do the final review of the papers.

We have a lot of customers and they have been struggling to adopt everything we have added to the language. We want to make sure that everything we add counts.

We should try to avoid single purpose solutions

When we look at the new proposal, and especially at the language side, they fall into 4 categories.

New features where we asked for papers. These are likely to succeed.

Improvements to existing features which help to generalise a feature. These are likely to succeed

Narrow scope features. We would like to ask incubators to take a look at those to see if there are any features that relate to each other.

Solo features which may be a good idea. They are unlikely to succeed unless they can be generalised.

Herb presents the WG21 groups.

Herb presents what days which group meets.

Herb presents the meeting schedule.

P1000 is the paper that will contain the progress of the standardisation.

This is the last meeting for new C++20 features, and features that need LWG changes

Anything in the incubators is not C++20 intended.

Mike Miller : I see Prague on the schedule of future meetings.

Herb : yes, the first meeting 2020 will be in Prague, we don't have any details yet.

When there are details, they will be posted on the isocpp page.

Herb continues.

You don't have to vote in the subgroups. If you haven't kept up with the discussion, it isn't helpful

Titus Winters : what about voting neutral ?

Herb : if you vote neutral , it means you know what is going on, but you have no opinion.

In subgroup anyone can vote, but please only vote if you have an informed opinion.

We will try to explicitly include the shipping vehicle in the subgroup polls.

Tom Honermann: how do we interpret consensus in polling ?

Herb : look at SD4 on standardisation page. Chair can decide what is consensus within their subgroup.

John : the link to SD4 is in the agenda. See "The WG21 Practices and Procedures, and Code of Conduct".

Herb continues : we will try to improve the queue of comments, to avoid out of order insertion and tangents.

Stuart Davis Herring : scribes, please write down to who something was a response.

Bryce Adelstein LeBach : it is important to capture the name . If you are taking the minutes, please ask for the name.

Herb : we always take notes. The scribe always gets to interrupt and make sure the name is correct.

Nathan Meyers : in WG sometimes people ask a question. Can they still ask them ?

Herb : what I said is a general rule, but chairs should take each case separately.

## 5. Plans for the future (special Monday session to discuss Cologne)

Plans for the future (PL22.16)

### 1. Next and following meetings

- 2019-02-18/23: Kona, HI, US ([N4765](#))
- 2019-07-15/20: Cologne, Germany ([N4780](#))

- 209-11-4/09 Belfast , Northern Ireland

How many people plan not to attend Kona : 15

How many people plan not to attend Cologne 17

How many people plan not to attend Belfast : 15

Nico Josuttis presents : Cologne meeting is not sponsored by a single company. Michael Spertus contributed too. If you plan to do the same, please contact me. There is an issue with the invitation, there will be an updated paper in the post San Diego mailing with a working link and other fixes. Please register before 10th of May. Any questions, please send me an email.

Jens Maurer presents : 14th of December is the registration cut off for Kona. Please book your hotel soon.

## 6. WG and SG sessions

John Spicer presents : during the week, CWG and LWG will be putting out proposal that will be voted on Saturday. Those need to be finalised and made available by 8pm PST Friday.

Only new papers go on the straw poll page, the older papers can be found in the mailing. If you have concerns, please make yourself known to chairs or on the mailings.

Jens Maurer presents meeting room schedule. The schedule can be found on the wiki.

Show of hands for each group to see if we need to amend the assigned rooms. Jens gives directions for each room.

SG5 will meet after the plenary meeting.

Lawrence Crowl : we will do issue triage today.

SG10 will meet during the week. See John Spicer.

Jens presents evening sessions.

Monday: Direction of metaprogramming(Ville Voutilainen).

Ville : This is an information-gathering session, for the chairs of EWG and LEWG to find out what the concerns with SG7's directional guidance are. It is not a debate. It is not a session that will take a poll. EWG and LEWG regulars are invited, as is Direction; the rest of the committee is uninvited. If you have concerns you wish to air, try to find an EWG or LEWG regular to air those concerns.

The duration of this session will be 60 minutes, after which further concerns will need to be conveyed offline.

Monday: Editorial session (Richard Smith) in room "Brittany". Schedule includes [structure.specifications] guidance.

Tuesday: Signedness and span (Bryce Adelstein LeBach)

Bryce : this is only an informative session, we will not be making any decisions.

Titus Winters: LEWG will have to schedule the actual vote for span's signedness, I haven't done that yet. I also encourage people to attend the evening session because the arguments have evolved a bit.

Wednesday: Education of C++ (Jan Christiaan van Winkel)

Thursday: SG15 Tooling (Titus Winters)

Thursday: LWG issues processing in room "Fairfield" (Marshall Clow)

Jens presents meeting times and the break times.

Breakfast should be available earlier than the meeting starts.

## 7. Review of the meeting (Saturday 8:30 AM)

John Spicer opened the meeting at 8:34am PST.

Herb Sutter presents.

Herb Sutter goes over the voting procedure.

Thank you to Michael Wong for chairing SG5. Michael will be stepping down and Hans Boehm has volunteered to take over as chair of SG5.

There are two new study groups :

SG19: Machine Learning (Michael Wong)

SG20: Education (JC van Winkel).

WG and SG status and progress reports.

- **SG5: Transactional memory (Wong)**

Michael Wong presents. There is a continuous discussion on reduced transactional memory in conjunction with the industry . Thank you to Hans for picking this up

- **SG6: Numerics (Crowl)**

Herb presents. SG6 met twice this week and proposed several papers.

- **SG7: Compile-time programming (Carruth)**

Chandler Carruth presents. Sg7 met for an afternoon and talked about 3 things.

Formal guidelines for what kinds of papers need review by sg7 before going to other groups, preview of what reflection facilities might look like if we build them on top of the new constexpr facilities, and new perspective on reflection on top of constexpr facilities, but from a different group. We will continue with both approaches so we can make a better decision on the direction we should take.

There was also an informational session in the evening to help spread information on compile time programming facilities added to constexpr.

- **SG10: Feature test (Nelson)**

John Spicer presents. SG10 is encouraging library and core group to check if a proposal should have a feature test macro. SG10 also went through some papers to see if they need FT macros. All decisions should be made through CWG and EWG as a part of the normal proposal handling.

- **SG12: Undefined and unspecified behavior (Dos Reis)**

Herb reads out report by Gabriel Dos Reis. SG12 had a joint session with EWG on Wednesday morning regarding UB and contracts. A partial resolution was sent to Core.

Thursday and Friday was a joint session with WG23. The group primarily recommended mitigation measures for vulnerabilities related to control structures. It is remarkable that those measures were already covered by the C++ Core Guidelines. Stephen Michell will make the new document available on WG23 website.

- **SG13: HMI & I/O (Human/Machine Interface) (Roger Orr)**

Roger Orr presents. The group is re-activating because of the P0267, the 2D graphics paper. SG13 voted to encourage the paper author to work with the contra papers (P1225R0 and P1062R0). The group also looked at web view paper (P1108R1) and encouraged further work. SG13 would like input from graphics experts but there is a problem of arranging a time to meet them.

- **SG14: Games & low latency (Wong)**

Michael Wong presents. SG14 had an impromptu meeting on linear algebra. There was a high level discussion of scope. SG15 also discussed frequency of meetings. Please contact Michael to ask about the meetings. Michael will assign chairs for each meeting.

SG14 plans to start having more face to face meetings. Guy Davidson, Ben Craig, and Ben Saks might take the lead on that.

SG14 will continue with two telecoms every month and triaging regular SG14 matters. Linear Algebra will be breaking up as a sub group/

Walter Brown : is numerics SG involved in linear algebra work ?

Micheal Wong : yes, and a few other SG have interest in linear algebra.

- **SG15: Tooling (Winters)**

Bryce Adelstein Leibach presents. SG15 had an evening session to discuss seven papers. There was a lot of good discussion. The direction of the proposals is not clear yet.

- **SG16: Unicode (Honermann)**

Tom Honermann presents. SG16 had first face to face meeting, it was well attended. SG16 talked about the direction and strategy, and identified a couple of priorities that the group will focus on. SG16 wants to improve support for transcoding SG16 looked at P1275 and worked on advancing on other papers, including some unicode features, mandating the encoding that's used for 16 and 32 integers, and normalisation.

SG16 looked at papers on efficient string building. The group also provided feedback to other SG on questions they had.

- **SG1: Concurrency (Giroux)**

Olivier Giroux presents. SG1 processed 34 paper, and talked about dismantling volatile.

Giroux presents SG1 delivery pipeline to C++20

jthread + interrupt\_token that supports callbacks, moved SG1 → LEWG.

Forward executors with one-way/one-way-bulk only, moved SG1 → LEWG.

The “C++20 Synchronization Library” omnibus paper moved LEWG → LWG.

Memory model repairs/improvements moving CWG → IS20.

No new/novel c++20 papers from now on. There is room for integration with existing features and fixes, any new work should be aimed at the next shipping vehicle.

Please don't rename concurrency term without consultation with SG1.

Co-routines: no SG1 issues; no library additions. Forward defects 31 & 32 to CWG. Additional library support for “awaitable” types needs more bake time in SG1.

Freestanding C++ direction needs clarifying. The interest among SG1 members is pretty substantial. It would be good if we could avoid disjointed polls.

Evening session summary in post mailing.

## ◦ **SG17: EWG Incubator (JF Bastien)**

Erich Keane presents

EWGI met all-day Monday / Tuesday / Wednesday.

These are the papers that EWG likely won't get to see in San Diego. EWGI will see the ones it can. We've prioritized papers that could plausibly make it to C++20 (either because they target a C++20 feature, or they're small). EWGI goal was first to disambiguate the shipping vehicle, and either forward to EWG for consideration this week (unlikely), or provide paper feedback.

EWGI saw 30 papers:

- 5 papers sent as-is to EWG
- 2 papers sent to EWG asking for feedback (will be seen again by EWGI)
- 6 papers had no consensus to move forward (we recorded objections, authors know what would get people to change their mind)
- Other papers got feedback, some need input from other groups, they'll be seen again by EWGI
- EWGI have a backlog of 27 papers
- EWGI spent most of Wednesday afternoon looking at the merged pattern matching proposal, and gave plenty of feedback to the authors

Papers we sent to EWG:

- [P1009R0](#) Array size deduction in new-expressions
- [P1041R1](#) Make char16\_t/char32\_t string literals be UTF-16/32
- [P1097R1](#) Named character escapes
- [P1099R2](#) Using Enum
- [P1301R0](#) nodiscard should have a reason: request that LEWG see this as well: maybe they want to add reasons to current nodiscard in the STL. Maybe EWG should see this after LEWG decides whether / when they want to use it.
- [P1203R0](#) Modular main(). Want EWG feedback, choose:
  1. Take the suggested direction
  2. Take the suggested direction and modify the signature of main (return type, return omission, argc/argv or other parameters, Unicode, environment, etc)
  3. Don't do modular main, instead add an **entrypoint** attribute
- [P1040R2](#) std::embed is forwarded to EWG not because it's ready, but because EWGI felt that EWG feedback was required on the approach:
  1. As proposed, allow a constexpr std::embed which takes a constexpr string (which can be created at compile-time!). There's worry that doing this breaks module's dependency scanning because we need to do full constexpr evaluation before discovering that a module depends on a file being embedded.
  2. Somehow use module syntax to import the file to embed, creating a constexpr datastructure. There's worry that this forbids generating the file name at compile-time.

Daveed Vandevorde : you mentioned p1009r0 is seen by core ?

Timur Doumler : no, it hasn't been seen yet.

## ◦ **Evolution (Voutilainen)**

Ville Voutilainen presents :

Concepts:

We reviewed [P1141R1](#), Yet another approach for constrained declarations, and approved it. Core moved it in San Diego. In addition, we approved [P0848R0](#), Conditionally Trivial Special Member Functions.

We did not look at [P0782R2](#), Constraining Concepts Overload Sets, despite it being on the agenda. This proposal changes dependent calls in constrained templates to consider only the overloads that were used to satisfy a constraint as viable candidates. We will possibly revisit the proposal in Kona, as we need to decide the fate of this change before C++20 ships.

Modules:

The merged Modules proposal, [P1103R1](#) was approved, with a bunch of fixes, including

- Making the module and import keywords context-sensitive
- Support for inline module partitions
- Change to when a module preamble is considered to end
- Fixes to redefinitions in legacy imports

The global module fragment was not removed, and the paper on retiring pernicious language constructs in module contexts was rejected.

The plan has been for quite some time to wait until Kona before moving Modules, so as to give implementers more time to gain implementation experience.

Coroutines:

We discussed [P1063R1](#), Core Coroutines. Evolution, again, still, has consensus (but far from unanimity) to move the merge of the Coroutines TS forward.

Aggregate initialization:

The proposal to allow paren-initializing aggregates, [P0960](#), was approved.

Contracts:

The change to allow preconditions and postconditions to use private members, [P1289R0](#), was approved.

We discussed UB and optimizations at quite some length; the work on that problem is ongoing, and we'll see more in Kona.

Some constexpr extensions

- [P1327](#), allowing `dynamic_cast` in constant expression was approved. It's also possible to use `typeid` and its comparison in constant expressions.
- We approved being able to change the active member of a union in constant expressions.
- We approved [P1073](#), "constexpr".

- We discussed "for..." and "for constexpr". Further work is expected with encouragement that we want facilities like that.

Template argument deduction:

We approved fixes that allow CTAD on aggregates, alias templates, and using inherited constructors.

All Your Spaceships Are Belong to Us:

The papers [P1185](#) and [P1186](#) were approved. We expect further discussion on spaceship in Kona.

There was no consensus on making comparison operators chain, nor was there encouragement for further work in that area.

Structured bindings:

We discussed removing the need for tuple\_element and allowing explicit typing of bindings. There was no consensus for either; Pattern Matching can solve the latter problem.

Defaulted special member functions and noexcept:

We approved [P1286](#), which changes noexcept on defaulted special member functions to no longer require the noexcept-spec to match what would've been implicitly generated.

Guaranteed constant initialization for non-constexpr variables:

We discussed [P1143](#), Adding the [[constinit]] attribute, and we expect to see a revision in Kona. The current guidance is for a keyword instead of an attribute.

Deprecations:

Deprecating volatile was discussed; a revised proposal is expected for Kona.

[P1161](#), Deprecate uses of the comma operator in subscripting expressions, was approved. This paves the way to multi-argument subscripting. We didn't approve making that possible yet, but the deprecation is approved.

## ◦ **SG18: LEWG Incubator (Bryce Adelstein Leibach)**

Bryce Adelstein Leibach presents: LEWGI met for 3.5 days, but we had a room for an extra morning so we processed some additional work. Our approach was :

direction : do we want to solve this problem

design : how should we solve this problem

describe: how do we specify the solution

The process was :

- overview of the proposal,

- directional discussion : do we want to solve this problem ?

- take a poll : do we want to spend more time on this proposal ?

If yes, then do some design discussion and design improvement until it is ready to go to the library group.

No consensus to spend more time: 10  
Sent to LEWG for C++20: 3  
Sent to LEWG for input: 2  
Sent to SG1 for Conc v2 TS: 1  
Ready for LEWG(I) design review for C++23: 10  
Further refinement in LEWGI needed: 6  
Total LEWGI papers seen: 32

C++20 papers (sent to LEWG)  
P1207R0 Moveability of Single-pass Iterator  
P1310R0 Unifying the Many Ways to Compare  
P0288R2 `std::unique_function`

Concurrency v2 TS papers  
P0876R3 `std::fiber_context`  
P0260R2 Concurrent Queues

LEWG Input Needed  
P1072R1 Optimized Initialization for `std::string`  
P1010R1 Container Support for Implicit Lifetime Types

Refine in LEWGI  
P0959R1 `std::uuid`  
P1275R0 `std::environment`  
P0447R4 `std::colony`  
P1276R0 Type Manipulation Facilities  
P1069R0 Refining Library CTAD Support  
P0638R0 Crochemore-Perrin `std::search`

C++23 (design review in LEWG(I))  
C++23 Smart Pointers:  
P0468R1 Intrusive Smart Pointers  
P1132R1 `std::out_ptr`  
P0211R0 `std::allocate_unique`  
P1066R1 `std::exception_ptr` inspection  
P0993R0  
P1066R1 `std::find_backwards`  
P1317R0 `std::apply` enhancements  
P1318R0  
P0205R0 Seeding with `std::random_device`

No Consensus to Spend More Time  
P0275R4 Dynamic Loading Library  
P1159R0 `std::any_iterator`  
P1201R0 `std::variant` Direct Comparisons  
P1196R0 Value-based `std::error_category` Comparison  
P1197R0 Non-allocating `std::error_category::message`  
P1198R0 `std::error_category::failed`

ABI break  
ABI break  
ABI break

P0343R0 High-order Metaprogramming Functions  
P1081R0 Empty Struct Types  
P0319R0 Emplace for `std::promise/std::future`

Ville : did you discuss ranges based solution for `find_backwards` algorithm ?

Bryce : no. We originally did, but the feeling in the room was that it wasn't important enough to prioritise for C++20

## ◦ **Library Evolution (Winters)**

Titus Winters presents. 64 papers reviewed (44 last time), 14 papers left unreviewed (~28 last time) of which ~5 that could potentially target C++20

Pablo Halpern : can you attribute the increase in productivity to anything ?

Titus: it is helpful to have the incubator group, we didn't look at things that weren't plausibly baked. It's also where we are in the cycle, a lot of things are just about done. The fact we panicked a little bit and may have pushed the schedule and discussion accordingly

Titus continues. Policy Changes discussed :

- Do we provide extensions by inspecting user-provided types?
- When do we mark constructors explicit?

Forwarded to LWG for IS

- P1085 Should Span Be Regular
- P1024 Usability Enhancements for `std::span`
- P0784 Standard containers and `constexpr`
- P0533 `constexpr` for `<cmath>` and `<cstdlib>`
- P1251 A more `constexpr` bitset
- P0595 `std::is_constant_evaluated()`
- P0340 Making `std::underlying_type` SFINAE-friendly
- P0920 Precalculated hash values in lookup
- P0627 Function to mark unreachable code
- P1165 Fixing allocator usage for `operator+(basic_string)`
- P1210 Completing the Rebase of Library Fundamentals, Version 3, Working Draft
- P0325 `to_array` from LFTS with updates
- P1083 Move `resource_adaptor` from Library TS to the C++ WP
- P1209 Adopt Consistent Container Erasure from Library Fundamentals 2 for C++20
- P1187 A type trait for `std::compare_3way()`'s type
- P1191 Adding `operator<=>` to types that are not currently comparable
- P1248 Fixing Relations
- P1295 Spaceship library update
- P1252 Ranges Design Cleanup
- P1243 Rangify New Algorithms

- P0645 Text Formatting
- P0881 A Proposal to add stack trace library
- P0883 Fixing Atomic Initialization
- P1233 Shift-by-negative in `shift_left` and `shift_right`
- P0549 Adjuncts to `std::hash`
- P1272 Byteswapping for `fun&&nuf`
- P0631 Math Constants
- P1280 Integer Width Literals
- P1186 When do you actually use `<=>`?
- P1135 The C++20 Synchronization Library
- P0201 `polymorphic_value`
- P1227 Signed `size()` functions
  - `span::size()` is unsigned, adds `std::ssize()`
  - Compatible with P1089

#### Discussed, Will Prioritize

- P1208 Adopt `source_location` from Library Fundamentals V3 for C++20
- P1293 `ostream_joiner`
- P1035 Input range adaptors
- P1206 Range constructors for standard containers and views
- P0798 Monadic operations for `std::optional`
- P0443 Unified Executors
- P0437 Numeric Traits for the Next Standard Library
- P0556 (revisited) Change `ceil2` input requirements
- P0586 Safe integral comparisons
- P1222 A Standard `flat_set`
- P0660 A Cooperatively Interruptible Joining Thread
- P1072 Optimized Initialization for `basic_string` and `vector`

#### Discussed but not forwarded

- P1259 Merge most of Networking TS into C++ Working Draft
- P0863 Fixing the `partial_order` comparison algorithm
- P1312 Comparison Concepts
- P1291 `std::ranges::less<>` Should Be More!
- P1255 A view of 0 or 1 elements: `view::maybe`
- P0943 Support C atomics in C++
- P1175 a simple and practical optional reference for C++

#### Discussed and rejected

- P0657 - Deprecate certain declarations in the global namespace
  - We are not aiming for user code to reclaim the global namespace
- P1182 - New names for the power-of-2 templates
- P0310 - Splitting node and array allocation in allocators
- P1163 Explicitly Implicifying explicit Constructors
- P0262 A Class for Status and Optional Value
- P0891 Let `strong_order` Truly Be a Customization Point!
  - LEWG would rather make types hashable than comparable

- **Core (Mike Miller)**

Mike Miller presents. CWG is preparing for c++20 and has looked at various papers that were forwarded from EWG. Total of 26 papers, 16 will be moved today including co routines. CWG looked at, but did not move today, modules paper. CWG need to refine that and do a review for Kona.

There are 4 issue resolutions that will be moved today. CWG reviewed additional number of issue resolutions at this meeting and 4 editorial issues that needed cwg input. Of the issues that were not approved, some were forwarded to EWG for input.

There will be a session after this meeting to process some remaining issues resolutions and prioritisation, and 3 telephone conferences which will take place first monday every month before the Kona meeting. CWG will be reviewing modules or doing issues processing at these telephone conferences.

There is a large backlog of issues resolutions where CWG determined a direction, but the drafting has not yet materialised. We would like to clear up that backlog.

## **CWG Motions**

### **Motion 1**

Move to accept as Defect Reports all issues in [P1350R0](#) (Core Language Working Group "tentatively ready" Issues for the November, 2018 (San Diego) meeting) and apply the proposed resolutions to the C++ working paper.

No discussion.

No objection to unanimous consent.

Motion passes.

### **Motion 2**

Move to apply the changes in the "Consolidated wording" section of [P0668R5](#) (Revising the C++ memory model) to the C++ working paper.

No discussion.

No objection to unanimous consent.

Motion passes.

### **Motion 3**

Move to apply the changes in [P0982R1](#) (Weaken Release Sequences) to the C++ working paper.

No discussion.

No objection to unanimous consent.

Motion passes.

#### **Motion 4**

Move to apply the changes in [P1084R2](#) (Today's *return-type-requirements* Are Insufficient) to the C++ working paper.

Mike Miller : there was an email just this morning to CWG that there is a conflict with another paper later on. We are aware of the conflict and the authors are in contact, so this shouldn't be a problem.

No discussion.

No objection to unanimous consent.

Motion passes.

#### **Motion 5**

Move to apply the changes in [P1131R2](#) (Core Issue 2292: *simple-template-id* is ambiguous between *class-name* and *type-name*) to the C++ working paper.

No discussion.

No objection to unanimous consent.

Motion passes.

#### **Motion 6**

Move to apply the changes in [P1289R1](#) (Access control in contract conditions) to the C++ working paper.

No discussion.

No objection to unanimous consent.

Motion passes.

#### **Motion 7**

Move to apply the changes in [N4775](#) (Working Draft, C++ Extensions for Coroutines) as modified by the changes in [P1356R0](#) (Coroutine TS ready issues (25 and 27)) to the C++ working paper.

No discussion.

There are objections in the room.

Herb Sutter: we had this vote in Rapperswill. This was proposed for c++17, then we decided to do a TS for alternative proposal to be brought forward. There weren't any then, but there are now. Here we get to say yes or no, but we don't decide on the design.

We get new Coroutines papers every meeting, so that's why they are put to vote again.

If this fails, we will probably see this again in Kona.

This is not just a repeat of rapperswill, there are also features from core coroutines proposal added to the current coroutines proposal.

Chandler : in Rapperswill there was an early presentation of core coroutines. It failed plenary. We decided to have it come back and get a better understanding. We had that. There was a much better presentation at this meeting. I don't think

there will be substantial increase of information before Kona. The big chunk of work is not one meeting away. I don't think there will be big change in current TS between now and Kona. We need to decide if we want co routines in C++20.

Fabio Fracassi : Is the merge supposed to be finished by c++20 ?

Chandler : the merge has not been discussed in Evolution.

Herb : we are talking two merges. The issues were merged, but the whole proposals were not merged.

Eric Niebler : noone is suggesting that a merge of the two proposals will be available in Kona.

Roger Orr : Bulgarian paper was a good response to current TS, and they have some serious concerns. We would like to look at those concerns and address them

Bjarne : I think we waited long enough, If we don't go the current way, we are looking at C++23 or even C++26. We have been waiting far too long

Herb : NBs are saying they want co routines in C++20. They also want to explore the other directions. We can't have both.

Herb polls the room

In favour : 34

Opposed : 17

Abstain : 10

Herb polls national bodies by roll call:

Bulgaria Opposed

Canada Opposed

Finland In favour

France In favour

Germany Opposed

Netherlands In favour

Poland Opposed

Russia In favour

Spain Abstain

Switzerland Opposed

UK Opposed

US In favour

Motion fails

Herb : NB approval has decreased compared to the similar straw poll during the previous meeting.

## **Motion 8**

Move to apply the changes in [P1002R1](#) (Try-catch blocks in constexpr functions) to the C++ working paper.

No discussion.  
No objection to unanimous consent.  
Motion passes.

#### **Motion 9**

Move to apply the changes in [P1327R1](#) (Allowing `dynamic_cast`, polymorphic `typeid` in Constant Expressions) to the C++ working paper.

No discussion.  
No objection to unanimous consent.  
Motion passes.

#### **Motion 10**

Move to apply the changes in [P1236R1](#) (Alternative Wording for [P0907R4](#) Signed Integers are Two's Complement) to the C++ working paper.

No discussion.  
No objection to unanimous consent.  
Motion passes.

#### **Motion 11**

Move to apply the changes in [P0482R6](#) (`char8_t`: A type for UTF-8 characters and strings (Revision 6)) to the C++ working paper.

No discussion.  
There are objections in the room  
In favour 37  
Opposed : 1  
Abstained : 17

Motion passes.

#### **Motion 12**

Move to apply the changes in [P1353R0](#) (Missing Feature Test Macros) to the C++ working paper.

No discussion.  
No objection to unanimous consent.  
Motion passes.

#### **Motion 13**

Move to apply the changes in [P1073R3](#) (Immediate functions) to the C++ working paper.

No discussion.  
No objection to unanimous consent.  
Motion passes.

#### **Motion 14**

Move to apply the changes in [P0595R2](#) (`std::is_constant_evaluated()`) to the C++ working paper.

No discussion.

No objection to unanimous consent.

Motion passes.

### **Motion 15**

Move to apply the changes in [P1141R2](#) (Yet another approach for constrained declarations) to the C++ working paper.

No discussion.

No objection to unanimous consent.

Motion passes.

### **Motion 16**

Move to apply the changes in [P1094R2](#) (Nested Inline Namespaces) to the C++ working paper.

No discussion.

No objection to unanimous consent.

Motion passes.

### **Motion 17**

Move to apply the changes in [P1330R0](#) (Changing the active member of a union inside constexpr) to the C++ working paper.

No discussion.

No objection to unanimous consent.

Motion passes.

Mike Miller : What do we do with the co routines issues that were ready ?

Herb : I'd rather not add more polls. Can we poll them in Kona ?

Mike : Yes

### ◦ **LWG (Clow)**

Marshal Clow presents.

- LWG saw about 45 papers
- Moving 25 - one quite large
- Small groups worked quite well for initial review
- Lots of work will be done on the reflector between now and Kona
  - Issue processing
  - Initial paper review
- LWG is planning on having two issue resolution telecons between now and Kona (probably one in December, one in January)

- LWG still has have a ton of work to do for C++20.
- A copy of these bits will be in an update to P0985 (LWG Chair post-meeting report) in the post-meeting mailing.

## **LWG Motions**

### **Concurrency TS**

#### **Motion 1**

Move to apply to the Concurrency TS working paper the proposed resolution of issue 2697 in [P1224R0](#) (C++ Standard Library Issues to be moved in San Diego).

Botond Ballo : is this for Concurrency TS v2 ?

John : yes

No objection to unanimous consent.

Motion passes.

### **Library Fundamentals TS**

#### **Motion 2**

Move to apply to the Library Fundamentals TS working paper the proposed resolution of issues 2960 and ~~3134~~ in [P1224R0](#) (C++ Standard Library Issues to be moved in San Diego).

Marshall: there were two issues in the pre meeting mailing, but we decided not to move 3134 here because the next motions covers that issue.

No discussion.

No objection to unanimous consent.

Motion passes.

#### **Motion 3**

Move to apply to the Library Fundamentals TS working paper the proposed wording in [P1210R0](#) (Completing the Rebase of Library Fundamentals, Version 3, Working Draft). This also resolves LWG issue 3134.

No discussion.

No objection to unanimous consent.

Motion passes.

### **Issues**

#### **Motion 4**

Move to apply to the C++ working paper the proposed resolutions of all of the issues except 2697, 2960 and 3134 in [P1224R0](#) (C++ Standard Library Issues to be moved in San Diego).

No discussion.

No objection to unanimous consent.

Motion passes.

## **Draft Standard**

### **Motion 5**

Move to apply to the C++ working paper the proposed wording in [P1123R0](#) (Editorial Guidance for merging P0019r8 and P0528r3).

No discussion.

No objection to unanimous consent.

Motion passes.

### **Motion 6**

Move to apply to the C++ working paper the proposed wording in [P0487R1](#) (Fixing operator>>(basic\_istream&, **CharT**\*) (LWG 2499)).

No discussion.

No objection to unanimous consent.

Motion passes.

### **Motion 7**

Move to apply to the C++ working paper the proposed wording in [P0602R4](#) (variant and optional should propagate copy/move triviality).

No discussion.

No objection to unanimous consent.

Motion passes.

### **Motion 8**

Move to apply to the C++ working paper the proposed wording in [P0655R1](#) (visit<R>: Explicit Return Type for visit).

No discussion.

No objection to unanimous consent.

Motion passes.

### **Motion 9**

Move to apply to the C++ working paper the proposed wording in [P0972R0](#) (<chrono> zero(), min(), and max() should be noexcept).

No discussion.

No objection to unanimous consent.

Motion passes.

### **Motion 10**

Move to apply to the C++ working paper the proposed wording in [P1006R1](#) (Constexpr in std::pointer\_traits).

No discussion.

No objection to unanimous consent.

Motion passes.

**Motion 11**

Move to apply to the C++ working paper the proposed wording in [P1032R1](#) (Misc constexpr bits).

No discussion.

No objection to unanimous consent.

Motion passes.

**Motion 12**

Move to apply to the C++ working paper the proposed wording in [P1148R0](#) (Cleaning up Clause 20).

No discussion.

No objection to unanimous consent.

Motion passes.

Marshall : LWG will meet after the plenary. We will do paper review or issues review, depending on the number of people. Poll for the number of people. Enough people to split into little groups.

Titus: LEWG will meet if we have enough people. Poll for people. There is a quorum.

Ville : EWG if we have enough people. Poll for people. There is a quorum.

**Motion 13**

Move to apply to the C++ working paper the proposed wording in [P0318R1](#) (unwrap\_ref\_decay and unwrap\_reference)

No discussion.

No objection to unanimous consent.

Motion passes.

**Motion 14**

Move to apply to the C++ working paper the proposed wording in [P0357R3](#) (reference\_wrapper for incomplete types)

No discussion.

No objection to unanimous consent.

Motion passes.

**Motion 15**

Move to apply to the C++ working paper the proposed wording in [P0608R3](#) (A sane variant converting constructor)

No discussion.  
No objection to unanimous consent.  
Motion passes.

#### **Motion 16**

Move to apply to the C++ working paper the proposed wording in [P0771R1](#)   
(`std::function` move constructor should be `noexcept`)

No discussion.  
No objection to unanimous consent.  
Motion passes.

#### **Motion 17**

Move to apply to the C++ working paper the proposed wording in [P1007R3](#)  
(`std::assume_aligned`)

No discussion.  
No objection to unanimous consent.  
Motion passes.

#### **Motion 18**

Move to apply to the C++ working paper the proposed wording in [P1020R1](#) (Smart pointer creation with default initialization)

No discussion.  
No objection to unanimous consent.  
Motion passes.

#### **Motion 19**

Move to apply to the C++ working paper the proposed wording in [P1285R0](#)  (Improving Completeness Requirements for Type Traits)

No discussion.  
No objection to unanimous consent.  
Motion passes.

#### **Motion 20**

Move to apply to the C++ working paper the proposed wording in [P1248R1](#) (Remove **CommonReference** requirement from **StrictWeakOrdering** (a.k.a Fixing Relations))

No discussion.  
No objection to unanimous consent.  
Motion passes.

#### **Motion 21**

Move to apply to the C++ working paper the proposed wording in [P0591R4](#) (Utility functions to implement uses-allocator construction)

No discussion.

No objection to unanimous consent.

Motion passes.

### **Motion 22**

Move to apply to the C++ working paper the proposed wording in [P0899R1](#) (LWG 3016 is Not a Defect)

No discussion.

No objection to unanimous consent.

Motion passes.

### **Motion 23**

Move to apply to the C++ working paper the proposed wording in [P1085R2](#)  (Should Span be Regular?)

No discussion.

There are objections in the room.

In favour : 30

Opposed : 1

Abstain : 25

Motion passes.

### **Motion 24**

Move to apply to the C++ working paper the proposed wording in [P1165R1](#) (Make stateful allocator propagation more consistent for `operator+(basic_string)`)

No discussion.

No objection to unanimous consent.

Motion passes.

### **Motion 25**

Move to apply to the C++ working paper the proposed wording in [P0896R4](#) (The One Ranges Proposal)

No discussion.

No objection to unanimous consent.

Motion passes.

### **Motion 26**

Move to apply to the C++ working paper the proposed wording in [P0356R5](#) (Simplified partial function application).

No discussion.

No objection to unanimous consent.  
Motion passes.

### **Motion 27**

Move to apply to the C++ working paper the proposed wording in [P0919R3](#) (Heterogeneous lookup for unordered containers).

No discussion.  
There are objections in the room.  
In favour : 26  
Opposed: 2  
Abstain: 25  
Motion passes

### **Motion 28**

Move to apply to the C++ working paper the proposed wording in [P1209R0](#) (Adopt Consistent Container Erasure from Library Fundamentals 2 for C++20).

No discussion.  
No objection to unanimous consent.  
Motion passes.

## ◦ **Direction Group**

Bjarne Stroustrup presents. For the new people, we have been around for about a year. The idea is to look at issues that go across the individual technical issues. We take input, but we can't do everything. If you have things that concern the language direction, contact us.

People ask for more detail of long-term direction. The original paper we wrote P0939Rx is focused on what would be a good direction on C++20, but gets more vague as we get into the future.

Now we are going to work on C++23. We never have clear straight forward answers. We say it would be a good idea to go in a certain direction.

People asked feedback on specific topics which indicates these are hot topics:

- signed/unsigned comparisons (and span)
- define what/who is the target of freestanding/embedded
- static reflection
- pattern matching
- ABI (what is breakage)
- is  $\Leftrightarrow$  really an improvement ?
- networking - is it right to delay past c++20 ? Why do we need more design review, when we had a design review a couple of years ago ?

Marshall: the review in Cologne wasn't really a design review, it was a discussion on how it would fit in the current library. It was a minor design review, but mostly a wording review.

Bjarne continues. If you have opinions about how something fits into the big picture, e-mail us, talk to use, we are not meant to be working in isolation. Thank you.

Herb : during the break there were more questions about co routines. We are likely to have another vote in Kona. There are NB's that said may change their vote to yes if they have more time. There will be papers for the next meeting, and they will go to an EWG schedule, and EWG can put the the poll forward again.

## 7. Closing activities

### 7.1 Issues delayed until today

No discussion.

### 7.2. PL22.16 motions, if any

John Spicer presents voting rules.

#### 1) **Approval of Systematic Review for ISO/IEC TS 19217:2015 -- C++ Extensions for Concepts**

Barry Hedquist moves.

Marshall Clow seconds.

In favour : 26 in the room + 3 by e-mail

Opposed : 0

Abstain : 0

Motion passes

Walter Brown presents.

Thank you the host and the sponsors.

Thank you to everyone that participated and those who helped us participate.

## 7.3 Mailings

- 2018-11-26: Post-San Diego
- 2019-01-21: Pre-Kona

Hal Finkel : web form to get paper number is going to stay up. Please use it over the next couple of weeks

Lawrence Crawl : where is the form ?

Hal : you will find it on the wiki page. You can also email me

## 8. Adjournment

Marshall Clow moves. Barry Hedquist seconds.

Approved by unanimous consent.

John Spicer adjourns the meeting at 11:01 am PST.

## 9. Attendance

| Name                   | Representing               | NB |
|------------------------|----------------------------|----|
| Aaron Ballman          | GrammaTech Inc             |    |
| ADAM David Alan Martin | MongoDB Inc                |    |
| Agustin Berge          | Louisiana State University |    |
| Alexandru Voicu        | AMD                        |    |
| Alisdair Meredith      | Bloomberg                  |    |
| Andrew Sutton          | University of Akron        |    |
| Andrew Turner          | Qualcomm Inc               |    |
| Anna Gringauze         | Microsoft Corporation      |    |
| Anton Polukhin         | Yandex.Taxi                | RU |
| Ashley Hedberg         | Google                     |    |

| Name                    | Representing                           | NB |
|-------------------------|----------------------------------------|----|
| Attila Feher            | Bloomberg                              |    |
| Barry Hedquist          | Perennial                              |    |
| Barry Revzin            | Jump Trading                           |    |
| Ben Craig               | National Instruments                   |    |
| Benjamin Saks           | Saks & Associates                      |    |
| Billy Baker             | FlightSafety International             |    |
| Billy R. O'Neal III     | Microsoft Corporation                  |    |
| Bjarne Stroustrup       | Morgan Stanley                         |    |
| Blaise Watson           |                                        |    |
| Bob Steagall            | KEWB Computing                         |    |
| Botond Ballo            | Mozilla                                | CA |
| Brian Van Straalen      | Lawrence Berkeley National Laboratory  |    |
| Bruno Lopes             | Apple                                  |    |
| Bryce Adelstein Leibach | NVIDIA Corporation                     |    |
| Carter Edwards          | NVIDIA Corporation                     |    |
| Casey Carter            | Microsoft Corporation                  |    |
| Chandler Carruth        | Google                                 |    |
| Chanyoung Park          |                                        | CA |
| Chris Kennelly          | Google                                 |    |
| Chris Mysen             | Google                                 |    |
| Christof Meerwald       | Programming Research Ltd               |    |
| Christopher Earl        | Lawrence Livermore National Laboratory |    |
| Corentin Jabot          |                                        | FR |
| Damien Lebrun-Grandie   | Oak Ridge National Laboratory          |    |

| Name                | Representing                   | NB |
|---------------------|--------------------------------|----|
| Dan Kalowsky        |                                |    |
| Daniel Sunderland   | Sandia National Laboratories   |    |
| Daveed Vandevoorde  | Edison Design Group            |    |
| David Goldblatt     | Facebook                       |    |
| David Hollman       | Sandia National Laboratories   |    |
| David Olsen         | NVidia Corporation             |    |
| David Sankel        | Bloomberg                      |    |
| David Stone         | Google                         |    |
| Davis Herring       | Los Alamos National Laboratory |    |
| Detlef Vollmann     | Vollmann Engineering           | CH |
| Dietmar Kühl        | Bloomberg                      |    |
| Dmitrii Kozhevnikov | Jetbrains                      |    |
| Eric Fiselier       | Google                         |    |
| Eric Niebler        | Facebook                       |    |
| Erich Keane         | Intel Corporation              |    |
| Fabio Fracassi      |                                | DE |
| Faisal Vali         |                                |    |
| Frank Birbacher     | Bloomberg                      |    |
| Gabriel Dos Reis    | Microsoft Corporation          |    |
| Gary Powell         | Sony Interactive Entertainment |    |
| Geoffrey Romer      | Google                         |    |
| Gevorg Voskanyan    |                                |    |
| Glen Fernandes      | C Plus Plus Alliance Inc       |    |
| Gor Nishanov        | Microsoft Corporation          |    |
| Graham Lopez        | Oak Ridge National Laboratory  |    |

| Name                  | Representing                    | NB |
|-----------------------|---------------------------------|----|
| Hal Finkel            | Argonne National Laboratory     |    |
| Hans Boehm            | Google                          |    |
| Herb Sutter           | Microsoft Corporation           |    |
| Howard Hinnant        | Ripple Labs                     |    |
| Hubert Tong           | IBM Corporation                 | CA |
| Hyrum Wright          | Google                          |    |
| Iain Sandoe           |                                 |    |
| J. Daniel García      | University Carlos III of Madrid | ES |
| J.C. van Winkel       |                                 | NL |
| James Bennett         |                                 |    |
| James Dennett         | Google                          |    |
| James Touton          | Blizzard                        |    |
| Jared Hoberock        | NVidia Corporation              |    |
| Jason Carey           | MongoDB Inc                     |    |
| Jason Merrill         | Red Hat Inc                     |    |
| Jason Rice            |                                 |    |
| Jean-Francois Bastien | Apple                           | CA |
| Jean-Paul Rigault     | Université de Nice S.A.         | FR |
| JeanHeyd Meneide      |                                 |    |
| Jeff Chapman          |                                 |    |
| Jeff Snyder           | PDT Partners                    | GB |
| Jeffrey Mendelson     | Bloomberg                       |    |
| Jennifer Yao          | Microsoft Corporation           |    |
| Jens Maurer           | Edison Design Group             |    |
| Jerry Coffin          |                                 |    |

| Name                | Representing                   | NB |
|---------------------|--------------------------------|----|
| John Lakos          | Bloomberg                      |    |
| John Spicer         | Edison Design Group            |    |
| Jon Kalb            | C Plus Plus Alliance Inc       |    |
| Jonathan Brian Coe  |                                | GB |
| Jonathan Caves      | Microsoft Corporation          |    |
| Jonathan Wakely     | Red Hat Inc                    | GB |
| Jorg Brown          |                                |    |
| Juan Alday          | GreenWireSoft                  |    |
| Kelly Walker        | Stellar Science                |    |
| Kirk Shoop          | Facebook                       |    |
| Lars Gullik Bjønnes | Cisco Systems Inc              |    |
| Lawrence Cowl       | Perennial                      |    |
| Lee Howes           | Facebook                       |    |
| Lewis Baker         | Facebook                       |    |
| Li-Ta Lo            | Los Alamos National Laboratory |    |
| Lisa Lippincott     | Tanium                         |    |
| Louis Dionne        | Apple                          | CA |
| Maged Michael       | Facebook                       |    |
| Mark Hoemmen        | Sandia National Laboratories   |    |
| Mark Zeren          | VMware Inc                     |    |
| Marshall Clow       | C Plus Plus Alliance Inc       |    |
| Mateusz Pusz        | EPAM Systems Inc               | PL |
| Mathias Stearn      | MongoDB Inc                    |    |
| Matt Calabrese      | Google                         |    |
| Matthew Woehlke     | Kitware, Inc                   |    |

| Name                | Representing                   | NB |
|---------------------|--------------------------------|----|
| Matthias Kretz      | GSI                            | DE |
| Mauro Bianco        | CSCS                           | CH |
| Michael Caisse      |                                |    |
| Michael Garland     | NVidia Corporation             |    |
| Michael McLaughlin  | Bob Taco Industries            |    |
| Michael Park        |                                |    |
| Michael Spencer     | Sony Interactive Entertainment |    |
| Michael Spertus     | Symantec                       |    |
| Michael Wong        | Codeplay                       | CA |
| Michael Young       |                                |    |
| Michał Dominiak     | NVidia Corporation             | PL |
| Mihail Mihailov     |                                |    |
| Minas Charalambides | UIUC                           |    |
| Muerte Isabella     | Target                         |    |
| Nathan Burgers      |                                |    |
| Nathan Myers        | Maystreet                      |    |
| Nathan Sidwell      | Facebook                       |    |
| Nathaniel Goodspeed | Linden Research, Inc           |    |
| Neal Meyer          | Amazon Corporate LLC           |    |
| Nevin Liber         | Ocient                         |    |
| Nicolai Josuttis    |                                | DE |
| Nicolas Lesser      |                                |    |
| Nina Dinka Ranns    | Edison Design Group            |    |
| Oleg Zhylin         |                                |    |
| Olivier Giroux      | NVidia Corporation             |    |

| Name              | Representing                           | NB |
|-------------------|----------------------------------------|----|
| P.J. Plauger      | Dinkumware Ltd                         |    |
| Pablo Halpern     | Intel Corporation                      |    |
| Paul McKenney     | IBM Corporation                        |    |
| Phil Nash         | Jetbrains                              |    |
| Rene Rivera       | C Plus Plus Alliance Inc               |    |
| Richard Smith     | Google                                 |    |
| Robert Douglas    | RKR Capital                            |    |
| Robert Romey      |                                        |    |
| Robert Simpson    | Qualcomm Inc                           |    |
| Roger Orr         |                                        | GB |
| Ronan Keryell     | Xilinx                                 |    |
| Ryan McDougall    | Zoox Inc                               | CA |
| S.W. Holeman II   |                                        |    |
| Sebastian Büttner |                                        |    |
| Sergei Murzin     | Bloomberg                              |    |
| Shuo Feng Liu     | IBM Corporation                        | CA |
| Sifang Li         | Hummingbird Code                       |    |
| Siu Chi Chan      | AMD                                    |    |
| Stephan Lavavej   | Microsoft Corporation                  |    |
| Stéphanie Even    | SILKAN                                 |    |
| Stephen Schurr    | Ripple Labs                            |    |
| Tana Plauger      | Dinkumware Ltd                         |    |
| Thomas Köppe      | Google                                 |    |
| Thomas Rodgers    | Red Hat Inc                            |    |
| Thomas Scogland   | Lawrence Livermore National Laboratory |    |

| Name              | Representing             | NB |
|-------------------|--------------------------|----|
| Tim Northover     | Apple                    |    |
| Tim Shen          | Google                   |    |
| Timur Doumler     | Jetbrains                | GB |
| Titus Winters     | Google                   |    |
| Tom Honermann     | Synopsys Inc             |    |
| Tomasz Kamiński   | Sabre                    | PL |
| Tony Van Eerd     | Christie Digital         | CA |
| Tyler Sutton      | Lock3 Software           |    |
| Victor Zverovich  | Facebook                 |    |
| Ville Voutilainen | Plum Hall Inc            | FI |
| Vincent Falco     | C Plus Plus Alliance Inc |    |
| Vincent Reverdy   | Paris Observatory        |    |
| Walter Brown      | Brown                    |    |
| Wesley Maness     | Schonfeld Tools LLC      |    |
| Will Deacon       | ARM Ltd                  | GB |
| William Miller    | Edison Design Group      |    |
| William Seymour   | Seymour                  |    |
| Wyatt Childers    |                          |    |
| Zach Laine        | Cadence                  |    |
| Zhihao Yuan       | DePaul University        |    |

